LOCAL STRUCTURE OF A FINITE TYPE QUASI-COHERENT SHEAF FLAT AT A POINT, BY ELEMENTARY ETALE NEIGHBOURHOOD AND OPEN SUBSCHEME

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file flat.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the chapter's own retained minimal preamble (sources/stacks-flat/retained-preamble.tex, lines 1-85) is a verbatim excerpt of the native Xy-pic setup and of the native theorem declarations and mathematical macros, followed by the project's own reference/slogan/history display-environment definitions copied mechanically from the shared preamble; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler, and extra_packages is left empty because everything needed is already in the retained preamble. No macro was invented or redefined.

Editorial note. Same-chapter pointers are located at their \label lines inside the pinned chapter file; cross-chapter pointers are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

Editorial note. The source gives the proposition two proofs, the first beginning at source line 2498 under the heading 'First proof' and saying that it is longer but does not use the existence of a complete devissage. The counted proof span is that first proof, which is a complete proof of the statement; the second proof immediately following was not taken, so that the counted pair stays one statement with one complete proof. The author's remark that the problem is local around the two points is used verbatim as the first reduction step.

Allowed standard prerequisites (editorial).

- Elementary devissage of a finite type quasi-coherent sheaf relative to a morphism. Same chapter flat.tex, label lemma-elementary-devissage, invoked at proof lines 2528 and 2550. Produces the elementary etale neighbourhoods, the diagram with Z' -> Y' -> S', the point z' and the finite type module G which is the starting point of the reduction.
- Descent of finite presentation, flatness and etaleness along the devissage. Same chapter flat.tex, labels lemma-devissage-finite-presentation (proof line 2552), lemma-devissage-flat (proof line 2553) and lemma-etale-at-point (proof line 2544). Shows the result for Y' -> S' with the pushed forward module implies the result for X -> S with F, and keeps the base change map etale after replacing S by a localisation.
- The induction step producing a finite free presentation with flat cokernel, and flatness and finite presentation in a short exact sequence. Same chapter flat.tex, label lemma-induction-step, invoked at proof line 2580; Algebra,

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labels algebra-lemma-finite-flat-local (proof line 2569), algebra-lemma-flat-ses (proof line 2611) and algebra-lemma-extension (proof line 2614); Descent, label descent-lemma-universally-injective-etale-open-immersion (proof line 2564). Gives the finite free module in the base case, supplies the map alpha with isomorphic reduction and flat cokernel in the induction step, and lets flatness and finite presentation be inferred for the middle term of the resulting short exact sequence.

Source reference. Original lemma 0519: sources/stacks-chow/tags.txt, source lines 6527–6527. Navigation pointer for \ref{algebra-lemma-extension} used inside the counted span (line 2614); the argument that consumes it is retained in the span.

Source reference. Original lemma 00NZ: sources/stacks-chow/tags.txt, source lines 877–877. Navigation pointer for \ref{algebra-lemma-finite-flat-local} used inside the counted span (line 2569); the argument that consumes it is retained in the span.

Source reference. Original lemma 00HM: sources/stacks-chow/tags.txt, source lines 648–648. Navigation pointer for \ref{algebra-lemma-flat-ses} used inside the counted span (line 2611); the argument that consumes it is retained in the span.

Source reference. Original lemma 02LC: sources/stacks-chow/tags.txt, source lines 3403–3403. Navigation pointer for \ref{descent-lemma-universally-injective-etale-open-immersion} used inside the counted span (line 2564); the argument that consumes it is retained in the span.

Source reference. Original lemma 057T: sources/stacks-flat/pinned-original-source.tex, source lines 598–598. Navigation pointer for \ref{lemma-devissage-finite-presentation} used inside the counted span (line 2552); the argument that consumes it is retained in the span.

Source reference. Original lemma 057U: sources/stacks-flat/pinned-original-source.tex, source lines 688–688. Navigation pointer for \ref{lemma-devissage-flat} used inside the counted span (line 2553); the argument that consumes it is retained in the span.

Source reference. Original lemma 057S: sources/stacks-flat/pinned-original-source.tex, source lines 494–494. Navigation pointer for \ref{lemma-elementary-devissage} used inside the counted span (line 2528); the argument that consumes it is retained in the span.

Source reference. Original lemma 05H2: sources/stacks-flat/pinned-original-source.tex, source lines 97–97. Navigation pointer for \ref{lemma-etale-at-point} used inside the counted span (line 2544); the argument that consumes it is retained in the span.

Source reference. Original lemma 05I3: sources/stacks-flat/pinned-original-source.tex, source lines 2356–2356. Navigation pointer for \ref{lemma-induction-step} used inside the counted span (line 2580); the argument that consumes it is retained in the span.

Proposition 0.1. *Let $f : X \to S$ be a morphism of schemes. Let $\mathcal{F}$ be a quasi-coherent sheaf on X. Let $x \in X$ with image $s \in S$. Assume that*

1. *f is locally of finite presentation,*
2. *$\mathcal{F}$ is of finite type, and*
3. *$\mathcal{F}$ is flat at x over S.*

Then there exists an elementary étale neighbourhood $(S', s') \to (S, s)$ and an open subscheme

$$V \subset X \times_S \mathrm{Spec}(\mathcal{O}_{S',s'})$$

which contains the unique point of $X \times_S \mathrm{Spec}(\mathcal{O}_{S',s'})$ mapping to x such that the pullback of $\mathcal{F}$ to V is an $\mathcal{O}_V$-module of finite presentation and flat over $\mathcal{O}_{S',s'}$.

First proof. This proof is longer but does not use the existence of a complete dévissage. The problem is local around x and s, hence we may assume that X and S are affine. During the proof we will finitely many times replace S by an elementary étale neighbourhood of (S, s). The goal is then to find (after such a replacement) an open $V \subset X \times_S \mathrm{Spec}(\mathcal{O}_{S,s})$ containing x such that $\mathcal{F}|_V$ is flat over S and finitely presented. Of course we may also replace S by $\mathrm{Spec}(\mathcal{O}_{S,s})$ at any point of the proof, i.e., we may assume S is a local scheme. We will prove the proposition by induction on the integer $n = \dim_x(\mathrm{Supp}(\mathcal{F}_s))$.

We can choose

(1) elementary étale neighbourhoods $g : (X', x') \to (X, x)$, $e : (S', s') \to (S, s)$,
(2) a commutative diagram

$$\begin{array}{ccccc} X & \xleftarrow{g} & X' & \xleftarrow{i} & Z' \\ & & & & \downarrow{\scriptstyle \pi} \\ {\scriptstyle f}\downarrow & & \downarrow & & Y' \\ & & & & \downarrow{\scriptstyle h} \\ S & \xleftarrow{e} & S' & = & S' \end{array}$$

(3) a point $z' \in Z'$ with $i(z') = x'$, $y' = \pi(z')$, $h(y') = s'$,
(4) a finite type quasi-coherent $\mathcal{O}_{Z'}$-module $\mathcal{G}$,

as in Lemma 057S. We are going to replace S by $\mathrm{Spec}(\mathcal{O}_{S',s'})$, see remarks in first paragraph of the proof. Consider the diagram

$$\begin{array}{ccccc} X_{\mathcal{O}_{S',s'}} & \xleftarrow{g} & X'_{\mathcal{O}_{S',s'}} & \xleftarrow{i} & Z'_{\mathcal{O}_{S',s'}} \\ & \searrow^{f} & \downarrow & & \downarrow{\scriptstyle \pi} \\ & & & & Y'_{\mathcal{O}_{S',s'}} \\ & & & \swarrow_{h} & \\ & & \mathrm{Spec}(\mathcal{O}_{S',s'}) & & \end{array}$$

Here we have base changed the schemes X', Z', Y' over S' via $\mathrm{Spec}(\mathcal{O}_{S',s'}) \to S'$ and the scheme X over S via $\mathrm{Spec}(\mathcal{O}_{S',s'}) \to S$. It is still the case that g is étale, see Lemma 05H2. After replacing X by $X_{\mathcal{O}_{S',s'}}$, X' by $X'_{\mathcal{O}_{S',s'}}$, Z' by $Z'_{\mathcal{O}_{S',s'}}$, and Y' by $Y'_{\mathcal{O}_{S',s'}}$ we may assume we have a diagram as Lemma 057S where in addition $S = S'$ is a local scheme with closed point s. By Lemmas 057T and 057U the result for $Y' \to S$, the sheaf $\pi_*\mathcal{G}$, and the point y' implies the result for $X \to S$, $\mathcal{F}$ and x. Hence we may assume that S is local and $X \to S$ is a smooth morphism of affines with geometrically irreducible fibres of dimension n.

The base case of the induction: $n = 0$. As $X \to S$ is smooth with geometrically irreducible fibres of dimension 0 we see that $X \to S$ is an open immersion, see Descent, Lemma 02LC. As S is local and the closed point is in the image of $X \to S$ we conclude that $X = S$. Thus we see that $\mathcal{F}$ corresponds to a finite flat $\mathcal{O}_{S,s}$ module. In this case the result follows from Algebra, Lemma 00NZ which tells us that $\mathcal{F}$ is in fact finite free.

The induction step. Assume the result holds whenever the dimension of the support in the closed fibre is $< n$. Write $S = \mathrm{Spec}(A)$, $X = \mathrm{Spec}(B)$ and $\mathcal{F} = \widetilde{N}$ for some

B-module N. Note that A is a local ring; denote its maximal ideal $\mathfrak{m}$. Then $\mathfrak{p} = \mathfrak{m}B$ is the unique minimal prime lying over $\mathfrak{m}$ as $X \to S$ has geometrically irreducible fibres. Finally, let $\mathfrak{q} \subset B$ be the prime corresponding to x. By Lemma 05I3 we can choose a map

$$\alpha : B^{\oplus r} \to N$$

such that $\kappa(\mathfrak{p})^{\oplus r} \to N \otimes_B \kappa(\mathfrak{p})$ is an isomorphism. Moreover, as $N_\mathfrak{q}$ is A-flat the lemma also shows that α is injective and that $\mathrm{Coker}(\alpha)_\mathfrak{q}$ is A-flat. Set $Q = \mathrm{Coker}(\alpha)$. Note that the support of $Q/\mathfrak{m}Q$ does not contain $\mathfrak{p}$. Hence it is certainly the case that $\dim_\mathfrak{q}(\mathrm{Supp}(Q/\mathfrak{m}Q)) < n$. Combining everything we know about Q we see that the induction hypothesis applies to Q. It follows that there exists an elementary étale morphism $(S', s) \to (S, s)$ such that the conclusion holds for $Q \otimes_A A'$ over $B \otimes_A A'$ where $A' = \mathcal{O}_{S',s'}$. After replacing A by A' we have an exact sequence

$$0 \to B^{\oplus r} \to N \to Q \to 0$$

(here we use that α is injective as mentioned above) of finite B-modules and we also get an element $g \in B$, $g \notin \mathfrak{q}$ such that Q_g is finitely presented over B_g and flat over A. Since localization is exact we see that

$$0 \to B_g^{\oplus r} \to N_g \to Q_g \to 0$$

is still exact. As B_g and Q_g are flat over A we conclude that N_g is flat over A, see Algebra, Lemma 00HM, and as B_g and Q_g are finitely presented over B_g the same holds for N_g, see Algebra, Lemma 0519. □